

Table 3: IoT applications in the public sector

Category	Sub-category	Example
Transportation	Public transportation	- Using GPS tracking devices, we can enable real-time monitoring of buses and give better information on waiting times - Using swipe-card information, we can analyse peaks in use and provide a better supply of buses
	Traffic	Using GPS tracking devices or presence detection sensors, we can enable real-time traffic analytics and have a smarter control of traffic lights to prevent traffic congestion
	Public bikes	Using GPS tracking devices, we can analyse where public bikes are most needed and also balance the availability of bikes at different locations
Economy	Tourism	Using beacons, we can push notifications on smartphones based on location and give people additional information about a place in a city or an art object in a museum
	Industry	Smart infrastructures
Health	Assistance to people	Using a single-push button, we can send an emergency alert
	Automated medication	Using medical devices, we can automate the supply of medicine for people with diabetes or high blood pressure and send alerts to the doctor when the device detects a problem
	Maintenance of medical devices and failure detection	Using a sensor, a battery-powered medical device could send a notification when a battery needs to be replaced
Environment	Water supply	Using sensors, leakage in water pipes can be detected
	Pollution	Using sensors, we can monitor pollution levels and the amount a polluting particle emitted
Administration	Electronic identity device	- Using magic bands, we can have all our identity information and important information updated - Using biometrics, personal magic bands can be secured and prevent identity theft
Security	Drones	Using computer vision, drones can identify security issues and report them
	Fire	- Using smoke detectors, automated notifications can be sent to intervention centres - Using sensors in gas pipes, we can identify potential sources of fire and deal with them before any damage is caused

Source: Deloitte, 2015

Challenges to be addressed

Even though the IoT is creating, and will likely continue to create, immense opportunities for both industry and consumers, it is not without its share of risks.

As is common when anything new comes onto the stage, legislators across the world are looking for ways to incorporate new regulatory frameworks to plug relevant holes of the IoT and ensure its smooth functioning. To realise the full potential of the IoT, businesses and governments need to work together to overcome the impediments that lay in its path.

Cyber security has already gained notoriety across the world due to

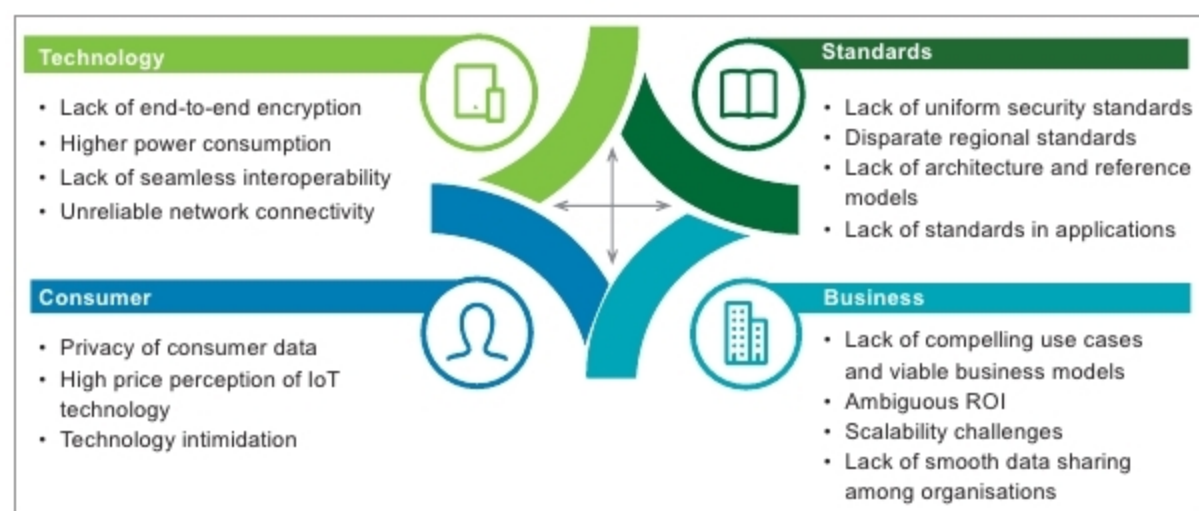


Figure 5: Key challenges impeding IoT growth and adoption (Source: Deloitte)

increased incidents of cyber-attacks, espionage and data breaches. With the number of connected devices increasing, and virtual and physical worlds seamlessly integrating, present security measures will no longer

remain adequate. Organisations will require enhanced security frameworks, ranging from device-level authentication and application security to system-wide assurance, resiliency and incidence response models.